

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 03 · LECTURE NOTES

Computing Equilibria

Existence without an algorithm, and a complexity class built for it.

MODULE OUTCOMES

- Explain why fixed-point existence gives no algorithm.
- Define PPAD and explain what makes it the right class.
- State the hardness of computing Nash equilibria.
- Explain what the hardness means for prediction.
- Identify the tractable special cases.

1 Existence without construction

NASH'S PROOF IS A FIXED-POINT ARGUMENT, WHICH ESTABLISHES EXISTENCE AND PROVIDES NO WAY TO FIND THE POINT

Brouwer's theorem says a continuous map from a ball to itself has a fixed point — and the standard proofs are non-constructive — which is a familiar situation in mathematics and a distinctly unfamiliar one in an algorithms course, where existence proofs are usually algorithms.

So 'an equilibrium exists' is entirely compatible with 'no efficient algorithm finds it' — and this course takes that gap seriously rather than treating existence as settling the matter, which is what distinguishes the algorithmic treatment from the classical one.

Which matters practically, not just aesthetically: if the agents cannot find the equilibrium, predicting that they will play it is unjustified — a prediction that requires solving an intractable problem is not a prediction, it is a description of a hypothetical.

And that is this module's real content — a computational critique of a solution concept — which is what algorithmic game theory contributed back to economics, and is a good example of a complexity result changing how a concept is used rather than only classifying it.

2 PPAD

THE PROBLEM WITH NP

NP-completeness is about problems where an answer might not exist. A Nash equilibrium ALWAYS exists, so "is there an equilibrium" is trivially yes -- there is nothing to decide, and the decision framing is useless.

THE IDEA

define a complexity class by the ARGUMENT that proves existence, rather than by the decision problem.

PPAD: Polynomial Parity Argument, Directed

the existence proof is: in a directed graph where every node has in-degree and out-degree at most one, if there is an unbalanced node then there must be another one.

FINDING the other one is the problem. The graph is exponentially large and given implicitly by a circuit.

AND NASH IS COMPLETE FOR IT

even for two players (Chen and Deng, 2006) which was the surprising part -- two-player zero-sum is in P, by linear programming

PPAD is defined by the existence argument rather than by the decision problem — which is why it had to be invented for this, and is a genuinely novel move in complexity theory (CSCE 637 Module 08's class zoo did not have a slot for it). **Two-player zero-sum is in P; general two-player Nash is PPAD-complete, and that contrast is the result**: the hardness does not come from many players, which is where everybody expected it.

AND PPAD-HARDNESS IS WEAKER EVIDENCE THAN NP-HARDNESS, WHICH IS WORTH SAYING

PPAD sits in territory where a polynomial algorithm would not collapse anything famous — so a polynomial algorithm for Nash would not imply $P = NP$, and the evidence of difficulty is correspondingly weaker than for an NP-hard problem.

But it is a well-studied class with many natural complete problems — Brouwer fixed points, Sperner's lemma, Arrow-Debreu market equilibria, and Nash — and none of them has yielded to decades of attention, which is the usual and perfectly respectable kind of evidence (CSCE 637 Module 01's framing of conditional hardness).

So the honest claim is: believed hard, on the same kind of evidence as most complexity conjectures — and the belief is well founded — which is a more careful statement than 'computing Nash equilibria is hard' and means the same thing to anyone who reads it carefully.

Which is CSCE 640 Module 12's proved/believed-with-evidence/assumed distinction arriving in a second setting — and the habit transfers exactly, which is part of why the two courses sit in the same semester.

3 What follows

- **A solution concept that cannot be computed is a weak predictor — if the analyst with a computer cannot find it, assuming the agents found it is not defensible**, and this is the argument that gave the hardness result its influence.
- **Which pushes attention toward dynamics (Module 12): what do agents who adapt over time actually reach, and how quickly?** — a question with better answers, and ones that do not require anybody to solve a hard problem.
- **And toward approximate equilibria**, in which no agent gains more than ϵ by deviating — **easier than exact, and still not known to be polynomial for small ϵ** , which is a sobering refinement of the bad news.
- **And toward designing games whose equilibria are easy to find — which is the designer's privilege**, since the designer chooses the game, **and is Module 06's entire approach**.
- **Which is the constructive reading of this module: if you are building the mechanism, build one whose equilibrium is obvious** — ideally one where each agent has a dominant strategy and does not need to reason about anybody else at all (Module 02 §2).

4 The tractable cases

TWO-PLAYER ZERO-SUM IS A LINEAR PROGRAM, AND THAT IS THE ONE CASE THAT IS GENUINELY EASY

Von Neumann's minimax theorem gives a well-defined value for any two-player zero-sum game, and the equilibrium strategies are the solution of a linear program — which is CSCE 669 Module 05's duality, and the two players' programs are duals of each other, which is where the minimax theorem comes from.

So zero-sum games are solved, completely and efficiently, for any size — **which is why they dominate the applications where this theory is actually deployed**: security games, some auction analyses, and the game-playing work in CSCE 625 and CSCE 642.

And potential games are the other good case (Module 05 §2): **best-response dynamics converge**, because every improving deviation decreases a single global potential function — which both guarantees a pure equilibrium exists and gives an algorithm that finds one.

Which is the practical summary: zero-sum and potential games are tractable, and general games are not — so check which kind you have before assuming an equilibrium will be reached, which is a two-minute check with a large payoff.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Daskalakis, Goldberg & Papadimitriou — The complexity of computing a Nash equilibrium (free)	§2 — the hardness result, with a simplified proof by the authors.
Papadimitriou — On the complexity of the parity argument (free)	§2 — where PPAD was defined, and the motivation is stated plainly.
Nisan et al., chapter 2 (free PDF)	§§1 to 3 — the computational treatment of equilibrium, including the algorithms that do work.
Roughgarden — lectures on complexity of equilibria (free)	§§3 and 4 — including the argument for why hardness matters for the concept's use.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State why a non-constructive existence proof is unsatisfying here.**
- 2 **Explain why NP-completeness is the wrong framing** for Nash.
- 3 **Describe the PPAD existence argument** in your own words.
- 4 **Name three other PPAD-complete problems.**
- 5 **Explain why PPAD-hardness is weaker evidence than NP-hardness.**
- 6 **Solve a zero-sum game by linear programming** and verify minimax.
- 7 **Show that the two players' programs are duals.**
- 8 **Find a potential function** for a small congestion game.
- 9 **Run best-response dynamics** on it and watch the potential decrease.
- 10 **Try the same on a general game** and find a cycle.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does Nash's proof give no algorithm?
- 2 Why does that matter for prediction?
- 3 Why is NP the wrong class for this problem?
- 4 What is PPAD defined by, and what is its existence argument?
- 5 What is complete for it, and what was surprising?

- 6 Why is PPAD-hardness weaker evidence, and is the belief still sound?
- 7 Give three consequences of the hardness.
- 8 What is the constructive reading?
- 9 Why is two-player zero-sum easy, and what does it reduce to?
- 10 What is the other tractable class, and why does it converge?